COMPETENCY STANDARD

FOR

GAS PIPE WELDING IN ENERGY SECTOR

(ENERGY SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

This document is available at:

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.
Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.

- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (20 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-BPI-GPW-01-G	Carry Out Workplace Interaction	1 Interpret workplace communication and etiquette. 2 Read and understand workplace documents. 3 Participate in workplace meetings and discussions. 4 Practice professional ethics at work	15
SICIP-BPI-GPW-02-G	Operate In a Team Environment	1 Identify team goals and work processes. 2 Identify own role and responsibilities within team. 3 Communicate and co-operate with team members. 4 Practice problem solving within the team.	15
		Total Hour	**30 Hrs.**

Sector Specific Competencies (10 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-BPI-GPW-01-S	Work with Hand Tools and Power Tools	1. Identify and inspect hand and power tools 2. Use hand tools properly and safely 3. Operate power tools properly and safely. 4. Clean and maintain hand and power tools	10
		Total Hour	**10 Hrs.**

Occupation Specific Competencies (320 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-BPI-GPW-01-O	Apply Safety, OHS, and Environmental Rules	1. Identify hazards and risks in pipeline welding. 2. Apply safety rules for hot work and confined spaces. 3. Use PPE and protective equipment correctly. 4. Follow environmental and emergency procedures.	10
SICIP-BPI-GPW-02-O	Interpret Pipeline Materials, Drawings, and Welding Requirements	1. Read and interpret working drawings. 2. Understand pipeline welding codes and ISO standards. 3. Interpret welding procedures. 4. Identify pipeline materials 5. Describe basic Arc and TIG welding principles. 6. Recognize welding machines and tools.	20
SICIP-BPI-GPW-03-O	Perform Arc Welding in 4G Position	1. Identify workpieces and materials for Arc welding. 2. Set Arc welding machine parameters for 4G position. 3. Execute Arc welding in 4G position. 4. Inspect weld bead quality and defects.	70
SICIP-BPI-GPW-04-O	Perform Pipe Fit-Up, Alignment, and Joint Preparation	1. Measure, cut and prepare pipes. 2. Bevel pipe ends and sets root face and root opening. 3. Align pipes using clamps, jigs, or supports. 4. Check alignment, root gap, and secure joints for welding.	20

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-BPI-GPW-05-O	Perform Arc Welding in 5G and 6G Position	1. Identify workpieces and materials for 5G and 6G. 2. Set Arc welding machine parameters for 5G and 6G positions. 3. Execute Arc welding in 5G and 6G positions. 4. Inspect weld bead quality and defects.	120
SICIP-BPI-GPW-06-O	Perform TIG welding in 5G and 6G Position	1. Identify workpieces and materials for 5G and 6G TIG welding. 2. Set TIG welding machine parameters for 5G and 6G positions. 3. Execute TIG welding in 5G and 6G position. 4. Inspect weld bead quality and correct defects.	80
		Total Hours	**0 Hrs.**

COMPETENCY STANDARD: GAS PIPE WELDING IN ENERGY SECTOR

Generic Competencies

Unit of Competency: CARRY OUT WORKPLACE INTERACTION	Nominal Duration: 15 hrs.	Unit Code: SICIP-BPI-GPW-01-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to carry out workplace interaction. It specifically includes the tasks of interpreting workplace communication and etiquette, reading and understanding workplace documents, participating in workplace meetings and discussions and practicing professional ethics at work.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret workplace communication and etiquette	1.1 Workplace codes of conduct are interpreted as per organizational guidelines. 1.2 Appropriate lines of communication are maintained with supervisors and colleagues. 1.3 Workplace interactions are conducted in a **<u>courteous manner</u>** to gather and convey information. 1.4 **<u>Workplace procedures and matters</u>** are comprehended.
2. Read and understand workplace documents	2.1 Workplace documents are interpreted correctly. 2.2 Visual information/symbols/signage are understood correctly and followed. 2.3 Specific and relevant information are accessed from **<u>appropriate sources</u>**. 2.4 Appropriate medium is used to transfer information and ideas.
3. Participate in workplace meetings and discussions	3.1 Team meetings are attended on time. 3.2 Meeting procedures and etiquette are followed. 3.3 Active participation is ensured, opinions are expressed and heard. 3.4 Inputs are provided and interpreted in line with the meeting purpose.
4. Practice professional ethics at work	4.1 Responsibilities as a team member are performed. 4.2 Tasks are performed in accordance with workplace procedures. 4.3 Confidentiality is maintained. 4.4 Inappropriate and conflicting situations are avoided.

Range of variables:

Variables	**Range** (may include but not limited to)

1. Courteous manner	1.1 Effective questioning 1.2 Active listening 1.3 Speaking skills 1.4 Writing skill 1.5 Email etiquette
2. Workplace procedures and matters	1.1 Notes 1.2 Arranging a meeting 1.3 Agenda 1.4 Simple reports such as progress and incident reports 1.5 Job sheets 1.6 Operational manuals 1.7 Brochures and promotional material 1.8 Visual and graphic materials 1.9 Standards 1.10 OHS information 1.11 Signs
3. Appropriate sources	1.1 Human Resources (HR) Department 1.2 Managers 1.3 Supervisors 1.4 Management Information System (MIS)

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace communication and etiquette 1.2 Workplace documents, signs and symbols 1.3 Meeting procedure and etiquette 1.4 Professional ethics
2. Underpinning skill	1.5 Demonstrating workplace communication and etiquette 1.6 Interpreting workplace instructions and symbols 1.7 Demonstrating active participation in workplace meeting 1.8 Applying professional ethics at work
3. Underpinning Attitudes	1.9 Commitment to occupational health and safety 1.10 Promptness in carrying out activities 1.11 Sincere and honest to duties 1.12 Environmental concerns 1.13 Eagerness to learn 1.14 Tidiness and timeliness 1.15 Respect for rights of peers and seniors in the workplace 1.16 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct

	4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Interpreted workplace communication and etiquette 1.2 Interpreted workplace instructions and symbols 1.3 Performed active participation in workplace meetings
2. Methods of assessment	Methods of assessment may include but not limited to: 1.4 Written test 1.5 Practical Demonstration 1.6 Oral Questioning 1.7 Portfolio (Optional)
3. Context of assessment	1.8 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 1.9 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **OPERATE IN A TEAM ENVIRONMENT**	**Nominal Duration:** 15 hrs.	**Unit Code:** SICIP-BPI-GPW-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate in a team environment. It specifically includes the tasks of identifying team goals and work processes, identifying own role and responsibilities within team, communicating and cooperating with team member, and practicing problem solving within the team.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1. Roles and objectives of the team are identified and interpreted. 1.2. Roles and responsibilities of team members and work processes are identified and interpreted.
2. Identify own role and responsibilities within team	2.1. Personal role and responsibilities are identified within the team environment. 2.2. Importance of relationships within and outside the team are interpreted .
3. Communicate and co-operate with team members	3.1. Other teammates' tasks are identified and support provided when requested. 3.2. The team is encouraged through **sharing information** or expertise, working together to solve problems, and putting team success first. 3.3. Views and opinions of other team members are interpreted and respected.
4. Practice problem solving within the team	4.1. Problems faced at the individual and team level are identified and showed insight into the root-causes of the problems. 4.2. A range of solutions and courses of action are identified together with benefits, costs, and risks associated with each. 4.3. The good ideas of others to help develop solutions are recognized and advice sought from those who have solved similar problems. 4.4. It is looked beyond the obvious and not stopped at the first answers.

Range of variables:

Variables	**Range** (may include but not limited to)

1. Sharing information	1.1. Agenda 1.2. Minutes 1.3. Progress and incident reports 1.4. Operational manuals 1.5. Visual and graphic materials 1.6. Emails and SMS 1.7. Policy, procedure and standards 1.8. OHS information

Curricular Content Guide

1. Underpinning knowledge	1.1. Team goals and work processes 1.2. Roles and responsibilities of the team 1.3. Finding problems and solving them 1.4. Importance of views and opinions of other team members
2. Underpinning skill	1.1. Identifying own role and responsibilities within team 1.2. Communicating and co-operating with team members 1.3. Demonstrating problem solving within the team
3. Underpinning Attitudes	1.4. Commitment to occupational health and safety 1.5. Promptness in carrying out activities 1.6. Sincere and honest to duties 1.7. Eagerness to learn 1.8. Tidiness and timeliness 1.9. Environmental concerns 1.10. Respect for rights of peers and seniors at workplace 1.11. Communication with peers and seniors at workplace
4. Resource implications	The following resources must be provided: 1.12. Workplace (simulated or actual) 1.13. Workplace documents, Projector 1.14. Learning manual 1.15. Tools, equipment and facilities appropriate to the process or activities. 1.16. Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 Identified own role and responsibilities within team 1.2 Communicated and co-operated with team members 1.3 Demonstrated problem solving within the team
2. Methods of assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Practical Demonstration

	2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: **WORK WITH HAND AND POWER TOOLS**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-BPI-GPW-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to use hand and power tools in the workplace. It specifically includes the task of identifying and inspecting hand and power tools, using hand tools properly and safely, operating power tools properly and safely and cleaning and maintaining hand and power tools.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify and inspect hand and power tools	1.1 Appropriate hand and power tools are identified. 1.2 Application of hand and power tools is recognized. 1.3 Usability of hand and power tools is checked and verified.
2. Use hand tools properly and safely	2.1. Appropriate **<u>hand tools</u>** are selected. 2.2. Safety precautions are ensured before using hand tools. 2.3. Unsafe or faulty hand tools are identified and marked for repair. 2.4. **<u>Measuring tools</u>** are checked and calibrated before use. 2.5. Use hand tools properly and safely to perform work activity.
3. Operate power tools properly and safely	3.1. Appropriate **<u>power tools</u>** are selected. 3.2. Power supply outlet and electrical cord are inspected and confirmed safe for use in accordance with established workplace safety requirements. 3.3. Safety precautions are ensured before using power tools in accordance with manufacturer's operating specification. 3.4. Proper sequence of operation applied for using power tools. 3.5. Unsafe or faulty power tools are identified and marked for repair. 3.6. Operate power tools properly and safely to perform work activity.
4. Clean and maintain hand and power tools	4.1. Dust and foreign matter is removed from hand and power tools in accordance to workplace standards. 4.2. Condition of hand and power tools is checked after use and reported.

	4.3. Appropriate lubricant is applied after use and prior to storage. 4.4. Defective hand and power tools are inspected and repaired or replaced. 4.5. Hand and power tools are stored and secured in accordance with workplace requirements.

Range of Variables

Variable	**Range** May include but not limited to:
1. Hand tools	1.1 Hammer 1.2 Bench vice 1.3 Files 1.4 Wrenches 1.5 Pliers 1.6 Scriber 1.7 Scraper Screw 1.8 Surface plate 1.9 Gauge 1.10 Tap sets 1.11 Die sets 1.12 Hacksaw 1.13 Spanners 1.14 Vice grip 1.15 Wire cutters 1.16 Clamps 1.17 Jacks
2. Power tools	2.1 Drills 2.2 Rivet gun 2.3 Grinders 2.4 Pneumatic wrenches 2.5 Press machine 2.6 Cutting 2.7 Saws 2.8 Soldering iron
3. Measuring tools	3.1 Micrometer 3.2 Megger 3.3 Measuring tape 3.4 Hose level 3.5 Water level 3.6 Calipers 3.7 Steel rule 3.8 Meter rule 3.9 Spirit level 3.10 Protractor 3.11 Tri-square

	3.12 Gauges

Curricular Content Guide

1. Underpinning Knowledge	1.1 Information on hand and power tools, their functions and use 1.2 Procedures for safely using hand and power tools
2. Underpinning Skills	2.1 Identifying hand, power and measuring tools 2.2 Following safety precautions when using hand, power and measuring tools 2.3 Using hand and measuring tools correctly and safely in accordance with manufacturer's operating specification 2.4 Operating power tools correctly and safely in accordance with manufacturer's operating specification 2.5 Cleaning and maintaining hand and power tools after use 2.6 Applying appropriate lubricant on hand and power tools after use and prior to storing
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Codes of conduct 4.5 Projector 4.6 Learning manual 4.7 Tools, equipment and facilities appropriate to the process or activities. 4.8 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and selected appropriate hand and power tools for work to be performed 1.2 identified and used measuring and testing tools appropriate to work activity 1.3 followed safety precautions when using hand and

	power tools 1.4 operated power tools safely and pursuant to manufacturer's operating specification 1.5 performed cleaning and maintenance of hand and power tools after use and prior to storing
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: APPLY SAFETY, OHS, AND ENVIRONMENTAL RULES	Nominal Duration: 10 Hrs.	Unit Code: SICIP-BPI-GPW-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply safety, ohs and environmental rules. It specifically includes the task of identifying hazards in pipeline welding, applying safety rules for hot work and confined spaces, using ppe and protective equipment correctly and following environmental and emergency procedures.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify hazards and risks in pipeline welding.	1.1 **Common hazards in pipeline welding** operations are identified and listed. 1.2 Environmental risks and waste materials are identified. 1.3 Findings from hazard and risk assessments are interpreted.
2. Apply safety rules for hot work and confined spaces.	2.1 Hot work permit requirements are reviewed and followed. 2.2 Confined space entry rules are applied according to workplace procedures. 2.3 Fire prevention measures are followed during welding operations. 2.4 Safe access, lighting, and communication procedures are maintained.
3. Use PPE and protective equipment correctly.	3.1 **Required PPE for pipeline welding** is identified and selected. 3.2 Protective equipment is worn correctly during all welding activities. 3.3 Respiratory protection is used where fumes or gases are present. 3.4 Damaged or defective PPE is reported and replaced.
4. Follow environmental and emergency procedures.	4.1 Environmental protection guidelines for welding operations are followed. 4.2 Waste materials, slag, and consumables are disposed of properly. 4.3 Emergency procedures for fire, gas leaks, and injuries are followed.

Range of Variables

Variable	Range (Includes but not limited to):
1. Common hazards in	1.1 Electrical hazards

pipeline welding	1.2 Fire and explosion hazards 1.3 Chemical and fume hazards 1.4 Thermal hazards (heat, burns) 1.5 Mechanical hazards 1.6 Ergonomic hazards 1.7 Noise hazards 1.8 Radiation hazards (arc flash/UV)
2. Required PPE for pipeline welding	2.1 Welding helmet 2.2 Safety goggles or face shield 2.3 Flame-resistant clothing (FR jacket/pants) 2.4 Welding gloves 2.5 Safety boots (steel toe, heat-resistant) 2.6 Respiratory protection 2.7 High-visibility vest (site requirement)

Curricular Content Guide

1. Underpinning Knowledge	1.1. Basic idea of hazards and risks involved in pipeline welding work. 1.2. Common welding hazards related to heat, sparks, fumes, gas, and electricity. 1.3. Environmental risks and types of waste produced during welding operations. 1.4. Purpose of hazard and risk assessment and how findings are understood. 1.5. Concept of hot work rules and permit requirements for safe welding. 1.6. Understanding of confined space dangers, access, ventilation, lighting, and communication needs. 1.7. Purpose and types of PPE required for pipeline welding activities. 1.8. Importance of respiratory protection when welding fumes or gases are present. 1.9. Knowledge of environmental protection measures and emergency actions for fire, gas leaks, and injuries.
2. Underpinning Skills	2.1. Identifying hazards and risks in pipeline welding operations. 2.2. Interpreting hazard and risk assessment findings. 2.3. Applying hot work and confined space safety rules. 2.4. Using PPE and protective equipment correctly. 2.5. Implementing fire prevention and safe work practices. 2.6. Following environmental protection procedures. 2.7. Responding to emergencies and reporting unsafe conditions.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties

	3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Tools, equipment and facilities appropriate to the process or activities. 4.2 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified hazards and risks in pipeline welding. 1.2 applied safety rules for hot work and confined spaces. 1.3 used PPE and protective equipment correctly. 1.4 followed environmental and emergency procedures.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **INTERPRET PIPELINE MATERIALS, DRAWINGS, AND WELDING REQUIREMENTS**	Nominal Duration: 20 Hrs.	Unit Code: SICIP-BPI-GPW-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to interpret pipeline materials, drawings, and welding requirements. It specifically includes the task of reading and interpreting working drawings. understanding pipeline welding codes and ISO standards, interpreting welding procedures, identifying pipeline materials, describing basic Arc and TIG welding principles and recognizing welding machines and tools.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Read and interpret working drawings.	1.1 **Working drawings** are understood for welding requirements. 1.2 Symbols, lines, and dimensions on drawings are interpreted. 1.3 Welding details are identified. 1.4 Material specifications and measurements are interpreted from drawings. 1.5 Drawing information is used to plan welding tasks.
2. Understand pipeline welding codes and ISO standards.	2.1 Applicable **pipeline welding codes and ISO standards** are identified. 2.2 Quality and safety guidelines from standards are interpreted. 2.3 Compliance requirements are identified for reference. 2.4 Standards information is applied during welding preparation.
3. Interpret welding procedures.	3.1 Approved Welding Procedure Specifications (WPS) are interpreted. 3.2 Joint preparation, fit-up, and welding sequence instructions are understood. 3.3 Procedure Qualification Records (PQR) documents are identified as needed. 3.4 Welder Qualification (WPQ) documents are identified as needed.
4. Identify pipeline materials	4.1 Types of pipeline materials are identified and listed. 4.2 Properties of carbon steel and alloy steel pipes are recognized. 4.3 Material markings and codes are interpreted. 4.4 Pipe thickness, diameter, and schedule are checked. 4.5 Material compatibility with welding processes is assessed.
5. Describe basic Arc	1.1 Basic principles of Arc and TIG welding processes are

and TIG welding principles.	explained. 1.2 Differences between Arc and TIG welding techniques are identified. 1.3 Heat generation, arc control, and shielding concepts are described. 1.4 Advantages and limitations of each process are reviewed.
6. Recognize welding machines and tools.	6.1 **<u>Types of welding machines</u>** used in pipeline welding are identified. 6.2 **<u>Basic parts and functions of welding machines</u>** are recognized. 6.3 **<u>Tools required for pipe cutting, beveling, and alignment</u>** are listed. 6.4 Machine settings and controls are interpreted for safe use.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Working drawings	1.1 Pipeline layout drawings 1.2 Welding procedure drawings (WPS details) 1.3 Isometric drawings 1.4 Piping assembly drawings 1.5 Joint detail drawings 1.6 Alignment drawings 1.7 Cross-section drawings 1.8 Site plan drawings
2. Pipeline welding codes and ISO standards	**Welding Codes** 2.1 API 1104 – Welding of Pipelines and Related Facilities 2.2 ASME B31.4 – Pipeline Transportation Systems for Liquids 2.3 ASME B31.8 – Gas Transmission and Distribution Piping Systems 2.4 AWS D1.1 – Structural Welding Code (used in many industrial projects) **ISO & QA/QC Standards** 2.5 ISO 3834 – Quality Requirements for Welding 2.6 ISO 9606 – Welder Qualification 2.7 ISO 15614 – Welding Procedure Qualification 2.8 ISO 14731 – Welding Coordination 2.9 ISO 9001 – Quality Management
3. Types of welding machines	3.1 AC Welding Machines 3.2 DC Welding Machines
4. Basic parts and functions of welding	**Parts**

machines	4.1 Electrode holder 4.2 Welding cables 4.3 Ground clamp 4.4 Control panel 4.5 Transformer/Inverter unit 4.6 Cooling fan 4.7 Gas regulator (for gas welding) **Functions** 4.8 Supply and control welding current and ampere 4.9 Hold and feed the electrode 4.10 Complete the electrical circuit 4.11 Regulate welding parameters 4.12 Provide cooling 4.13 Deliver shielding gas (if applicable)
5. Tools required for pipe cutting, beveling, and alignment	**For Cutting tools** 5.1 Pipe cutter 5.2 Cutting torch 5.3 Magnetic pipe cutter 5.4 Angle grinder 5.5 Bandsaw **For Beveling tools** 5.6 Pipe beveling machine 5.7 Grinder with beveling disc 5.8 Hand file **For Alignment tools** 5.9 Pipe alignment clamp 5.10 Pipe stands 5.11 Spirit level 5.12 Measuring tape 5.13 Square (T-square)

Curricular Content Guide

1. Underpinning Knowledge	1.1. Basic understanding of working drawings used for pipeline welding. 1.2. Meaning of symbols, lines, dimensions, and welding details shown on drawings. 1.3. Purpose of using drawing information to plan welding tasks correctly. 1.4. Awareness of common pipeline welding codes and

	ISO standards and why they matter. 1.5. Understanding of Welding Procedure Specifications (WPS), PQR, and WPQ documents. 1.6. Knowledge of common pipeline materials and basic properties of carbon steel and alloy steel. 1.7. Interpretation of material markings, pipe size, thickness, and schedule. 1.8. Basic principles of Arc welding and TIG welding and how they differ. 1.9. Recognition of welding machines, their main parts, controls, and common welding tools used in pipeline work.
2. Underpinning Skills	2.1. Reading and interpreting working drawings and welding symbols. 2.2. Understanding and applying pipeline welding codes and ISO standards. 2.3. Interpreting Welding Procedure Specifications (WPS) and related documents. 2.4. Identifying pipeline materials, specifications, and markings. 2.5. Assessing material compatibility with welding processes. 2.6. Understanding basic Arc and TIG welding principles. 2.7. Recognizing welding machines, tools, and interpreting machine controls.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Tools, equipment and facilities appropriate to the process or activities. 4.2 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 read and interpreted working drawings. 1.2 understood pipeline welding codes and ISO standards. 1.3 interpreted welding procedures. 1.4 identified pipeline materials

	1.5 described basic Arc and TIG welding principles. 1.6 recognized welding machines and tools.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM ARC WELDING IN 4G POSITION**	Nominal Duration: 70 Hrs.	Unit Code: SICIP-BPI-GPW-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform Arc welding in 4G position. It specifically includes the task of iidentifying workpieces and materials for Arc welding. setting Arc welding machine parameters for 4G position, executing Arc welding in 4G position and inspecting weld bead quality and defects.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify workpieces and materials for Arc welding.	1.1 Safety precautions are followed and **<u>PPEs</u>** are used. 1.2 Welding procedure specifications (WPS) for 4G welding are reviewed and understood. 1.3 3/8” or 1” base plate are collected as per WPS. 1.4 Dimensions, thickness, and joint configuration of workpieces are measured and confirmed as per 4G welding. 1.5 Workpieces are marked and positioned in the overhead position.
2. Set Arc welding machine parameters for 4G position.	2.1 Welding procedure specifications for 4G position are understood. 2.2 Current, voltage, and polarity settings are adjusted as required. 2.3 Electrode type and size are selected according to job specifications. 2.4 Machine output is tested to ensure stable arc performance. 2.5 Trial welds are performed to confirm parameter suitability.
3. Execute Arc welding in 4G position.	3.1 Workpiece is positioned and secured for the 4G welding setup. 3.2 Electrode angle, arc length, and travel speed are maintained correctly. 3.3 Root, fill, and cap passes are deposited with proper fusion. 3.4 Slag is removed and inter-pass cleaning is performed as required. 3.5 Heat input is controlled to avoid distortion and defects.
4. Inspect weld bead quality and defects.	4.1 Measurements of weld bead size, reinforcement, and profile are taken using appropriate gauges and tools. 4.2 Applicable **<u>non-destructive testing (NDT) methods</u>** are conducted to detect subsurface defects.

	4.3 Any weld defects or irregularities are identified and classified according to severity. 4.4 The weld bead is verified against welding procedure specifications (WPS).

Range of Variables

Variable	**Range** (Includes but not limited to):
1. PPEs	1.1 Safety glasses 1.2 Ear plugs 1.3 Gloves 1.4 Apron 1.5 Helmet 1.6 Mask 1.7 Safety shoes
2. Non-destructive testing (NDT) methods	2.1 Dye penetrant 2.2 Magnetic particle 2.3 Ultrasonic testing 2.4 Radiograph

Curricular Content Guide

1. Underpinning Knowledge	1.1. Basic understanding of workpieces and base materials used for Arc welding in the 4G (overhead) position. 1.2. Purpose and key information contained in Welding Procedure Specifications (WPS) for 4G welding. 1.3. Knowledge of plate thickness, dimensions, joint configuration, and correct overhead positioning of workpieces. 1.4. Understanding of Arc welding machine parameters. 1.5. Recognition of electrode types and sizes suitable for overhead Arc welding. 1.6. Basic principles of Arc welding in the 4G position. 1.7. Understanding of weld pass sequence such as root, fill, and cap and the importance of inter-pass cleaning. 1.8. Awareness of common weld bead characteristics.
2. Underpinning Skills	2.1. Identifying and preparing workpieces and materials according to WPS. 2.2. Applying safety precautions and using PPE correctly. 2.3. Setting and adjusting Arc welding machine parameters for 4G position. 2.4. Selecting correct electrodes and controlling arc characteristics. 2.5. Performing Arc welding in the 4G position with proper technique. 2.6. Controlling heat input and maintaining weld quality during passes. 2.7. Inspecting weld beads and identifying defects using

	appropriate methods.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Tools, equipment and facilities appropriate to the process or activities. 4.2 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified workpieces and materials for Arc welding 1.2 set arc welding machine parameters for 4G position 1.3 executed Arc welding in 4G position 1.4 inspected weld bead quality and defects
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM PIPE FIT-UP, ALIGNMENT, AND JOINT PREPARATION**	**Nominal Duration:** 20 Hrs**.**	**Unit Code:** SICIP-BPI-GPW-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform pipe fit-up, alignment, and joint preparation. It specifically includes the tasks of measuring, cutting and preparing pipes, beveling pipe ends and sets root face and root opening, aligning pipes using clamps, jigs, or supports and checking alignment, root gap, and secure joints for welding.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Measure, cut and prepare pipes	1.1 Job specifications and drawings are reviewed and understood. 1.2 **<u>Pipe dimensions</u>** are measured accurately using appropriate tools. 1.3 Cutting lines and markings are made according to requirements. 1.4 **<u>Pipes</u>** are cut and beveled following approved procedures. 1.5 Edges and surfaces are cleaned and prepared for welding. 1.6 Prepared pipes are inspected to ensure they meet job standards.
2. Bevel pipe ends and sets root face and root opening.	2.1 Bevel angles are marked according to job specifications. 2.2 Pipe ends are beveled using approved cutting or grinding tools. 2.3 Root face is set to the required thickness. 2.4 Root opening is adjusted to meet welding standards. 2.5 Beveled edges are cleaned and checked for defects.
3. Align pipes using clamps, jigs, or supports.	3.1 **<u>Alignment tools and supports</u>** are selected as required. 3.2 Pipes are positioned according to job specifications. 3.3 Alignment tools and supports are applied to hold pipes in correct alignment. 3.4 Root gap and internal alignment are checked and adjusted. 3.5 Misalignment is corrected before tack welding.
4. Check alignment, root gap, and secure joints for welding.	4.1 Pipe alignment is inspected against job specifications. 4.2 Root gap is measured and adjusted as required. 4.3 Internal and external fit-up conditions are verified. 4.4 Joints are secured using clamps or tack welds. 4.5 Fit-up quality is checked for uniformity and stability.

Range of Variables

Variable	Range (Includes but not limited to):
1. Pipe dimensions	1.1 Outside diameter (OD) 1.1.1 6” to 8” 1.2 Inside diameter (ID) 1.2.1 6” to 8” 1.3 Wall thickness 1.4 Pipe schedule 1.5 Pipe length 1.6 Pipe grade 1.7 Pipe weight (per meter)
2. Pipes	2.1 Carbon steel pipe 2.2 Stainless steel pipe 2.3 Alloy steel pipe 2.4 Seamless pipe 2.5 ERW (Electric Resistance Welded) pipe 2.6 Spiral welded pipe 2.7 API-grade line pipe (like API 5L)
3. Alignment tools and supports	3.1 Pipe alignment clamps 3.2 Chain clamps 3.3 Line-up clamps 3.4 Pipe stands 3.5 Rollers 3.6 Jack stands 3.7 Spirit level 3.8 Wedges and spacers

Curricular Content Guide

1. Underpinning Knowledge	1.1. Understanding of job specifications and drawings used for pipe cutting and preparation work. 1.2. Knowledge of pipe measurement methods and tools. 1.3. Concept of marking, cutting, and beveling pipes. 1.4. Purpose of cleaning pipe edges and surfaces to ensure weld quality. 1.5. Understanding of bevel angle, root face, and root opening requirements for pipeline welding. 1.6. Knowledge of common tools and methods used for pipe beveling and edge preparation. 1.7. Concept of pipe alignment and the role of clamps, jigs, and supports in maintaining correct position. 1.8. Importance of correct root gap, internal alignment, and fit-up before welding. 1.9. Understanding of joint stability and fit-up quality.
2. Underpinning Skills	2.1. Reading and interpreting job specifications and drawings.

	2.2. Measuring, marking, cutting, and preparing pipes accurately. 2.3. Beveling pipe ends and setting correct root face and root opening. 2.4. Selecting and using clamps, jigs, and alignment tools. 2.5. Aligning pipes and correcting misalignment before welding. 2.6. Checking root gap, fit-up, and joint stability. 2.7. Inspecting prepared joints to ensure compliance with job standards.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Tools, equipment and facilities appropriate to the process or activities. 4.2 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 measured, cut and prepared pipes. 1.2 beveled pipe ends and sets root face and root opening. 1.3 aligned pipes using clamps, jigs, or supports. 1.4 checked alignment, root gap, and secure joints for welding
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM ARC WELDING IN 5G AND 6G POSITION**	**Nominal Duration:** 120 Hrs**.**	**Unit Code:** SICIP-BPI-GPW-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform Arc welding in 5G and 6G position. It specifically includes the tasks of identifying workpieces and materials for 5G and 6G, setting Arc welding machine parameters for 5G and 6G positions. executing Arc welding in 5G and 6G positions and inspecting weld bead quality and defects.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Identify workpieces and materials for 5G and 6G.	1.1 Safety precautions are followed and **<u>PPEs</u>** are used. 1.2 Relevant welding procedure specifications (WPS) for 5G and 6G welding are understood. 1.3 Fixed 6” schedule 40 pipe are identified as per 5G and 6G welding requirement. 1.4 Workpieces are cleaned, beveled, and prepared as required. 1.5 Workpieces are arranged and clamped in the horizontal and inclined 45-degree fixed position. 1.6 Alignment, gap, and fit-up between workpieces are checked and adjusted to meet welding standards.
2. Set Arc welding machine parameters for 5G and 6G positions.	2.1 Welding procedure specifications for 5G and 6G are reviewed. 2.2 Current, voltage, and polarity settings are adjusted according to requirements. 2.3 Electrode type and size are selected as per pipe material and position. 2.4 Machine output is tested to ensure stable arc performance. 2.5 Trial welds are performed to confirm parameter suitability.
3. Execute Arc welding in 5G and 6G positions.	3.1 Workpieces are positioned and secured for 5G and 6G welding setups. 3.2 Electrode angle, arc length, and travel speed are maintained correctly. 3.3 Root, fill, and cap passes are deposited with proper fusion and penetration. 3.4 Slag is removed and inter-pass cleaning is performed as required. 3.5 Heat input is controlled to prevent distortion and welding defects.
4. Inspect weld bead	4.1 Measurements of weld bead size, reinforcement, and

quality and defects.	profile are taken using appropriate gauges and tools. 4.2 Applicable **non-destructive testing (NDT) methods** are conducted to detect subsurface defects. 4.3 Any weld defects or irregularities are identified and classified according to severity. 4.4 The weld bead is verified against welding procedure specifications (WPS).

Range of Variables

Variable	**Range** (Includes but not limited to):
1. PPEs	1.1 Safety glasses 1.2 Ear plugs 1.3 Gloves 1.4 Apron 1.5 Helmet 1.6 Mask 1.7 Safety shoes
2. Non-destructive testing (NDT) methods	2.1 Dye penetrant 2.2 Magnetic particle 2.3 Ultrasonic testing 2.4 Radiograph

Curricular Content Guide

1. Underpinning Knowledge	1.1. Basic understanding of workpieces and pipe materials used for 5G and 6G Arc welding positions. 1.2. Purpose and key requirements of Welding Procedure Specifications (WPS) for 5G and 6G welding. 1.3. Concept of cleaning, beveling, and preparing pipe ends to meet welding standards. 1.4. Understanding of alignment, root gap, and fit-up requirements before welding fixed pipes. 1.5. Basic idea of Arc welding machine parameters. 1.6. Recognition of suitable electrode types and sizes for different pipe materials and welding positions. 1.7. Understanding of welding pass sequence-root, fill, and cap. 1.8. Knowledge of weld bead inspection, common weld defects.
2. Underpinning Skills	2.1 Identifying and preparing pipes and workpieces for 5G and 6G welding. 2.2 Applying safety precautions and using PPE correctly. 2.3 Interpreting WPS for fixed pipe welding positions. 2.4 Setting and adjusting Arc welding machine parameters for 5G and 6G. 2.5 Executing Arc welding with correct technique in fixed positions.

	2.6 Controlling heat input and maintaining proper fusion and penetration. 2.7 Inspecting weld beads and identifying defects against WPS requirements.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Tools, equipment and facilities appropriate to the process or activities. 4.2 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified workpieces and materials for 5G and 6G. 1.2 set Arc welding machine parameters for 5G and 6G positions. 1.3 executed Arc welding in 5G and 6G positions. 1.4 inspected weld bead quality and defects.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM TIG WELDING IN 5G AND 6G POSITION**	**Nominal Duration:** 80 Hrs**.**	**Unit Code:** SICIP-BPI-GPW-06-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to perform TIG welding in 5G and 6G position. It specifically includes the tasks of Identifying workpieces and materials for 5G and 6G TIG welding, setting TIG welding machine parameters for 5G and 6G positions. executing TIG welding in 5G and 6G position and inspecting weld bead quality and correct defects.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify workpieces and materials for 5G and 6G TIG welding.	1.1 **<u>Workpieces</u>** required for 5G and 6G welding are identified from job specifications. 1.2 Pipe material type, grade, and thickness are verified. 1.3 **<u>Joint design</u>** and bevel preparation requirements are reviewed. 1.4 Filler rod type and size are selected according to material specifications. 1.5 **<u>Surface condition</u>** and cleanliness of workpieces are inspected. 1.6 Material compatibility for TIG welding is confirmed.
2. Set TIG welding machine parameters for 5G and 6G positions.	2.1 Welding procedure specifications for 5G and 6G positions are reviewed. 2.2 Current, voltage, and polarity settings are adjusted according to requirements. 2.3 Shielding gas type and flow rate are set as specified. 2.4 Tungsten electrode type, size, and tip preparation are selected and checked. 2.5 Filler rod selection is verified for material compatibility. 2.6 Trial welds are performed to confirm parameter accuracy before welding.
3. Execute TIG welding in 5G and 6G position.	3.1 Workpieces are positioned and secured for 5G and 6G welding setups. 3.2 Torch angle, arc length, and travel speed are maintained correctly. 3.3 Root, fill, and cap passes are deposited with proper penetration and fusion. 3.4 Filler rod is fed smoothly and consistently according to position requirements. 3.5 Inter-pass cleaning is performed to maintain weld quality.
4. Inspect weld bead	4.1 Weld beads are visually inspected for uniformity,

quality and correct defects.	penetration, and fusion. 4.2 Surface **defects** are identified. 4.3 Dimensional accuracy and weld profile are checked against specifications. 4.4 Defective areas are marked for corrective action. 4.5 Repairs are performed using appropriate TIG welding techniques.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Workpieces	1.1 Pipe type 1.2 Pipe size 1.3 Pipe schedule 1.4 Weld position (5G/6G)
2. Joint design	2.1 Joint type 2.2 Bevel type 2.3 Bevel angle 2.4 Root gap 2.5 Root face
3. Surface condition	3.1 Rust 3.2 Oil/grease 3.3 Paint 3.4 Moisture 3.5 Cleanliness level
4. Defects	4.1 Cracks 4.2 Porosity 4.3 Undercut 4.4 Overlap 4.5 Lack penetration 4.6 Excess penetration 4.7 Slag inclusion

Curricular Content Guide

1. Underpinning Knowledge	1.1. Understanding of workpieces and pipe materials used for 5G and 6G TIG welding. 1.2. Knowledge of pipe material type, grade, thickness, and joint design requirements. 1.3. Concept of bevel preparation and surface cleanliness needed for quality TIG welding. 1.4. Understanding of Welding Procedure Specifications (WPS) for 5G and 6G TIG welding. 1.5. Basic knowledge of TIG welding machine settings. 1.6. Understanding of weld pass sequence-root, fill, and cap-and the need for inter-pass cleaning. 1.7. Knowledge of weld bead inspection, common surface

	defects, dimensional checks, and corrective actions.
2. Underpinning Skills	2.1. Identifying and preparing workpieces and materials for 5G and 6G TIG welding. 2.2. Verifying material type, grade, thickness, and joint preparation requirements. 2.3. Interpreting WPS and setting TIG welding machine parameters correctly. 2.4. Selecting shielding gas, tungsten electrodes, and filler rods appropriately. 2.5. Performing TIG welding in fixed 5G and 6G positions with correct technique. 2.6. Controlling heat input, arc length, and filler rod feeding. 2.7. Inspecting weld quality and correcting defects as required.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Tools, equipment and facilities appropriate to the process or activities. 4.2 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified workpieces and materials for 5G and 6G TIG welding. 1.2 set TIG welding machine parameters for 5G and 6G positions. 1.3 executed TIG welding in 5G and 6G position. 1.4 inspected weld bead quality and correct defects.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.

	3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	**Gas Pipe Welding in Energy Sector**
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sq) OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

Sl. No.	Major Equipment and Training facilities	Required facilities
1.	Computers/ Laptop	01
2.	Multimedia Projector	01
3.	Internet connectivity	01
4.	AC/DC Arc welding machine	10
5.	TIG welding machine	10
6.	Gas cutting set	03
7.	Spare cylinder set (Argon, oxygen and acetylene)	01
8.	Hand grinding machine	10
9.	Bench vice	10
10.	Tool box set for welding workshop	10
11.	Fire extinguisher	04
12.	Welding booth with exhaust system	10
13.	First aid box	03

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Gas Pipe Welding in Energy Sector the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.